

The Grammar Matrix: Modules

Delph-in Meeting

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Previous versions of the Matrix

- April 2002: First version, from grammars for English, Japanese
- June 2002: Improved support for lexical rules
- March 2003: Standardized notation for semantic composition of MRS
- August 2003: Further refinement of architecture for MRS, RMRS
- October 2003: Bug fixes in rules, based on first uses of Matrix
- April 2004: Added lexical types
- September 2004: Added inventory of HEAD types

Modules: Dimensions of cross-linguistic variation

Extensions to language-independent core grammar

- Typologically motivated
- Auto-generated from data gathered via web interface
- Customizable: Human-readable (.tdl) files
- Scalable to broad coverage

Two kinds of typological variation

- Systems realized differently in each language

Basic word order: VSO, SVO, ...

Sentence negation: modifier, affixation, ...

Yes/no questions: sentence particle, word order, ...

- Phenomena found in only some languages

Noun incorporation

Numeral classifiers

Auxiliary verbs

Prototype Matrix modules

- Language-specific choices gathered via HTML form
 - Parameter setting
 - Affix and lexical item orthography
- Script generates .tdl files to extend Matrix grammar
- Resulting grammar delivered ready to parse/generate

Basic word order

- Specializations of head-subject and head-complement types
- Tuning to deal with spurious ambiguity when free word order
- Prototype variations:
SOV, SVO, VSO, VOS, OVS, OSV, V-final, V-initial, free
- Further work:
V2, main/subordinate variation, complement order variation

Yes-no questions

- Inversion of verb and subject relative to basic order
 - Main verbs
 - Auxiliary verbs
- Sentence particles
 - S-initial
 - S-final
- Further work:
 - Punctuation, intonation contrasts

Sentence negation

- Verbal inflection
 - Main or auxiliary or either
 - Prefix or suffix
- Independent word
 - Modifier (of V, VP, S)
 - Complement (of main, aux, or either)
- Mixed strategy: five possible interactions
- Further work:
 - Non-concatenative morphology

Constraints on design of modules

- Interface questions must be intelligible to working linguist
- Resulting grammar must be readable and extensible
- Efficient re-use of types and constraints

Potential consumers

- Pedagogy - grammar writers, field linguists
- Language documentation
- Re-use of resources for minority languages
- Typological study

Making Matrix Grammars Useful

- SEM-I: documentation of semantic expressivity
- Standardized MRS test suites
- Stochastic models for disambiguation
- Support for hybrid deep/shallow processing
- MRS-to-MRS mappings
- Efficient generation